DE–3600

Sub. Code
13

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

CLASSICAL ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Show that $\underset{n\to\infty}{lt}\ n^{1/n} = 1$. (10)

   (b) State and prove D'Alembert's ratio test. (10)

2. (a) Test the convergence of the series $\sum \frac{1}{n(\log n)^p}$. (10)

   (b) Discuss the convergence of the series $\sum (-1)^n \frac{1}{n^p}$. (10)

3. (a) Show that $\sum_0^\infty \frac{5n+1}{(2n+1)!} = \frac{e}{2} + \frac{2}{e}$. (10)

   (b) Prove that $\frac{5}{1\cdot 2\cdot 3} + \frac{7}{3\cdot 4\cdot 5} + \frac{9}{5\cdot 6\cdot 7} + \ldots = 3\log 2 - 1$. (10)

4. (a) Sum to infinity the series

$$\frac{15}{16}+\frac{15\cdot 21}{16\cdot 24}+\frac{15\cdot 21\cdot 27}{16\cdot 24\cdot 32}+\ldots\ldots \quad (10)$$

(b) Evaluate $\sum_{1}^{\infty}\frac{n^3+1}{(n+1)!}$. (10)

5. (a) Solve the equation $x^4+4x^3+5x^2+2x-2=0$ of which one root is $-1+i$. (10)

(b) If α, β, γ are the roots of $x^3+px^2+qx+r=0$, then find $(1+\alpha^2)(1+\beta^2)(1+\gamma^2)$. (10)

6. (a) If the roots of the equation $x^3+px^2+qx+r=0$ are in arithmetic progression, show that $2p^3-9pq+27r=0$. (10)

(b) Find s_{16} for the equation $x^8+ax+b=0$. (10)

7. (a) If a, b, c are positive and not all equal, prove that $(a+b+c)(ab+bc+ca)>9abc$. (10)

(b) Find the inverse of $\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$. (10)

8. (a) Determine whether the following equations are consistent and solve them if they are consistent. $3x-y+2z=1, x-y+z=-1, 2x-2y+3z=2$. (10)

(b) Find the eigen values and eigen vectors of $\begin{bmatrix}2 & 2 & 1\\ 1 & 3 & 1\\ 1 & 2 & 2\end{bmatrix}$. (10)

DE–3601

Sub. Code
14

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

(5 × 20 = 100)

1. (a) (i) Find $\frac{dy}{dx}$ if $x^m y^n = (x+y)^{m+n}$.

(ii) Find $\frac{dy}{dx}$ if $x^3 + y^3 = 3axy$.

(b) (i) If $y = \left[x + \sqrt{1+x^2}\right]^m$ prove that $(1+x^2)y_2 + xy_1 - m^2 y = 0$.

(ii) Find $\frac{d^2y}{dx^2} y = 5x^3 + 7x^2 + 4x + 8$.

2. (a) Find the equation of the tangent and normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $(a\cos\theta, b\sin\theta)$.

(b) Find the envelope of the family of straight lines $\frac{x}{a} + \frac{y}{b} = 1$ where $a^2 + b^2 = k^2$.

3. (a) (i) Prove that

$$\int \sqrt{a^2 - x^2}\,dx = x/2\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}(x/a) + c$$

(ii) Evaluate $\int \frac{dx}{e^x + e^{-x}}$. (12)

(b) Show that $\int_0^{\pi/2} \sin^3 x \cos^2 x dx = 2/15$. (8)

4. (a) Obtain a reduction formula for $\int x^m (\log x)^n\, dx$ and apply to evaluate $\int_0^1 x^4 (\log x)^3\, dx$.

(b) If $I_n = \int_0^{\pi/2} x^n \sin x\, dx$ prove that

$$I_n + n(n-1)\, I_{n-2} = n(\pi/2)^{n-1}.$$

5. (a) Solve $\frac{dy}{dx} + y\cot x = 4x\operatorname{cosec} x$ given that $y = 0$ when $x = \frac{\pi}{2}$.

(b) Solve $(2x+1)2y'' - 2(2x+1)y' - 12y = 6x$.

6. (a) Solve $y'' + 4y' + 13y = 2e^{-x}$ given $y(0) = 0$ and $y'(0) = 1$.

(b) Solve $(D^2 - 4)y = \sin^2 3x$.

7. (a) Find $L\left[x^2 \cosh ax\right]$

(b) Find $L^{-1}\left[\dfrac{1}{(s+1)(s^2+1)}\right]$.

8. (a) (i) Solve $(y-z)p+(z-x)q = x-y$

(ii) Solve $yzp + zxq = xy$. (12)

(b) Solve $Z = px + qy + (q/p) - p$. (8)

DE–3602

Sub. Code
15

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

ANALYTICAL GEOMETRY AND VECTOR CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Find λ so that the equation $12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$ represents a pair of straight lines. Also, find the separate equations of the lines and their point of intersection. (12)

   (b) Show that the circles $x^2 + y^2 + 2x - 8y + 8 = 0$ and $x^2 + y^2 + 10x - 2y + 22 = 0$ touch each other. (8)

2. (a) Find the equation of the circle which bisecks the circumference of the circles $x^2 + y^2 = 1$; $x^2 + y^2 + 2x - 3 = 0$ and $x^2 + 2y - 3 = 0$.

   (b) Find the Cartesian equation equivalent of the polar equation $r\cos\theta = 4$.

3. (a) Find the equation of the plane which passes through the points $(2, 1, -2)$ $(-1, 4, 1)$ and $(1, 1, 1)$.

   (b) Find the equation of the plane which passes through the point $(1, 1, -7)$ and $(2, 1, -9)$ and is perpendicular to the plane $x - 2y + 2z + 5 = 0$.

4. (a) Find the image of the line,

$\frac{x-1}{1}=\frac{y-3}{-2}=\frac{z-4}{-3}$ in the plane $2x-y+z+3=0$.

(b) Find the angle between the line,

$\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$ and the plane $ax+by+cz+d=0$.

5. (a) Find the equation of the sphere passing through the points $(1,1,-2);(-1,1,2)$ and whose center is on the line $x+y-2-1=0=2x-y+z-2$.

(b) Find the equation of the common perpendicular to the lines

$\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}$; $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}$.

6. (a) Find the equation of the sphere passing through the points $(2,0,1)$ $(0,-2,3)(4,-1,2)$ and $(1,-5,-1)$.

(b) Show that the plane $2x-2y+z+12=0$ touches the sphere $x^2+y^2+z^2-2x-4y+2z-3=0$ and find the point of contact.

7. (a) Prove that $Curl(r\times a)=-2a$ where 'a' is a constant vector.

(b) If $\bar{r}=x\vec{i}+y\vec{j}+z\vec{k}$ prove that

(i) $\nabla r^n=n\cdot r^{n-2}\cdot r$

(ii) Find grad ϕ given $\phi(x\,y\,z)=x^2y+y^3z$.

8. Verify Gauss divergence theorem for

$f=(2xy-z)\vec{i}+x^2y\vec{j}-(x+3)\vec{k}$ over the region bounded by $2x+2y+z=6$; $x=0$; $y=0$; and $z=0$.

DE–3603

Sub. Code
23

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

MECHANICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) State parallelogram of forces and give an analytical expression for the resultant of two forces acting at a point. (8)

   (b) Two beads of weights w and w' can slide on a smooth circular wire in a vertical plane. They are connected by a light string which subtends an angle 2β at the centre of the circle when the beads are in equilibrium on the upper half of the wire. Prove that the inclination of the string to the horizontal is given by $\tan\alpha = \frac{w \sim w'}{w + w'}\tan\beta$. (12)

2. (a) OA, OB, OC are the lines of action of two forces P and Q and their resultant R respectively. Any transversal meets the line in L, M and N respectively. Prove that $\frac{P}{OL} + \frac{Q}{OM} = \frac{R}{ON}$. (10)

(b) $ABCDEF$ is a regular hexagon and at A, act forces represented by $\overline{AB}, 2\overline{AC}, 3\overline{AD}, 4\overline{AE}$ and $5\overline{AF}$. Show that the magnitude of the resultant is $AB\sqrt{351}$ and that it makes an angle $\tan^{-1}\left(\frac{7}{\sqrt{3}}\right)$ with AB. (10)

3. (a) Find the resultant of two unlike and unequal parallel forces acting on a rigid body. (8)

(b) State and prove Varigon's theorem of moments. (12)

4. (a) If three coplanar forces acting on a rigid body keep it in equilibrium, then prove that they must either be concurrent or be all parallel. (10)

(b) State five laws of friction. (10)

5. (a) Show that the path of a projectile is a parabola. (10)

(b) A particle is thrown over a triangle from one end a horizontal base and grazing the vertex falls on the other end of the base. If A, B are the base angles, and α the angle of projection, show that $\tan\alpha = \tan A + \tan B$. (10)

6. (a) If a perfectly elastic sphere impinges on a fixed smooth plane, then prove that its velocity is not altered by impact and the angle of reflection is equal to the angle of incidence. (12)

(b) Find the loss of kinetic energy due to direct impact of two smooth spheres. (8)

7. (a) If the displacement of a moving point at any time be given by an equation of the form $x = a\cos wt + b\sin wt$, show that the motion is a simple Harmonic motion. (10)

(b) Find the velocity and acceleration in polar coordinates. (10)

8. (a) Find the differential equation of central orbits. (10)

(b) In every central orbit, prove that the a real velocity is constant and the linear velocity varies inversely as the perpendicular from the centre upon the tangent to the path. (10)

DE–3604

Sub. Code
24

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Let M be any non-empty set and $x, y \in M$. Define $d(x,y) = \begin{cases} 0 & if \quad x = y \\ 1 & if \quad x \neq y \end{cases}$. Then prove that d is a metric on M. (10)

   (b) Prove that any non-empty open subset of $\mathbb{R}$ can be expressed as a countable union of mutually disjoint open intervals. (10)

2. (a) Prove that A is closed if and only if A contains all its limit points. (10)

   (b) Prove that f is continuous if and only if inverse image of closed set is closed. (10)

3. (a) Prove that every, convergent sequence is a Cauchy sequence in a metric space. (10)

   (b) Prove that l_p is a complete metric space. (10)

4. (a) State and prove Cantor's intersection theorem. (10)

   (b) Prove that $\mathbb{R}$ is of second category. (10)

5. (a) State and prove Heine Borel theorem. (10)

   (b) Let A be a subset of a metric space M. If A is totally bounded then prove that A is bounded. (10)

6. (a) Prove that a metric space (M,d) is totally bounded if and only if every sequence in M has a Cauchy subsequence. (10)

   (b) Prove that continuous image of a compact metric space is compact. (10)

7. (a) Give an example of a function which is convergence but not uniform. (10)

   (b) State and prove Cauchy criterion for uniform convergence. (10)

8. (a) State and prove contraction mapping theorem. (10)

   (b) State and prove Piccard's theorem. (10)

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DE–3605

Sub. Code
25

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

PROBABILITY AND STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) If A and B are any two events in a sample space, then prove that

(i) $P(A \cap B) \geq 1 - P(\overline{A}) - P(\overline{B})$

(ii) $P(\bigcup_i Ai) \leq \sum_i P(Ai)$

(iii) $P(\bigcap_{i=1}^{\infty} Ai) \geq 1 - \sum_{i=1}^{\infty} P(Ai)$

(b) (i) If A, B, C are mutually independent events, then prove that $A \cup B$ and C are independent events.

(ii) State and prove Baye's theorem.

2. (a) If X is continuous random variable with p.d.f given by $f(x) = \begin{cases} kx & 0 \leq x \leq 2 \\ 2k & 2 \leq x \leq 4 \\ -kx + 6k & 4 \leq x \leq 6 \\ 0 & \text{elsewhere} \end{cases}$

Then find the following :

(i) K

(ii) $P(X \le 3)$

(iii) $P(0 \le X \le 4)$

(iv) $P(X \ge 2)$

(v) $F(X)$

(b) Find the

(i) Mean

(ii) Median

(iii) Mode for the following distribution

$$f(x) = \begin{cases} 6(x - x^2), & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

3. (a) Fit a straight line to the following data and estimate the value of Y corresponding to $X = 6$.

x	0	5	10	15	20	25
y	12	15	17	22	24	30

(b) If the correlation coefficient of X and Y exists, show that $-1 \le P \le 1$.

4. (a) Discuss area property of normal distribution.

(b) If X is normally distributed with mean 8 and standard deviation 4 find

(i) $P(5 \le X \le 10)$

(ii) $(10 \le X \le 15)$

(iii) $P(X \ge 15)$

(iv) $P(X \le 5)$

5. (a) X is a random variable following binomial distribution with mean 2.4 and variance 1.44. Find $P(x \geq 5)$ and $P(1 < X \leq 4)$.

(b) If the probability of success is $\frac{1}{100}$, how many trials are necessary in order that probability of atleast one success is grater than $1/2$.

6. (a) Five coins are tossed 320 times. The number of heads observed is given below. Examine whether the coin is unbiased.

No. of heads	0	1	2	3	4	5	Total
Frequency	15	45	85	95	60	26	320

(b) Analyses the variance in the following Latin square

A8	C18	B9
C9	B18	A16
B11	A10	C20

7. Derive Fisher's F distribution.

8. (a) Fit a Poisson distribution for the following data and test the goodness of fit.

x :	0	1	2	3	4	5	6	Total
f :	273	70	30	7	7	2	1	390

(b) Given the following positive class frequencies. Find the remaining class frequencies $N = 20$; $(A) = 9$; $(B) = 12$; $(C) = 8$; $(AB) = 6$; $(BC) = 4$; $(CA) = 4$; $(ABC) = 3$.

DE–3606

Sub. Code
31

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) If A, B and C are three sets then prove that (10)

   (i) $A \Delta B = (A \cup B) - (A \cap B)$

   (ii) $A \Delta (B \Delta C) = (A \Delta B) \Delta C$.

   (b) Let $f : X \to Y$ be a function. If $A \subseteq X$ and $B \subseteq Y$. Show that (10)

   (i) $A \subseteq f^{-1}[f(A)]$

   (ii) $f[f^{-1}(B)] \subseteq B$

   (iii) Give an example to show that equality need not hold in (i) and (ii)

   (iv) In each case state when will the equality hold?

2. (a) If A, B and C are any three finite sets prove that

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$$

(8)

(b) Let $f : A \to B$, $g : B \to C$ be bijections. Then prove that $g \circ f : A \to C$ is also a bijection. (6)

(c) State and prove Demorgan's laws. (6)

3. (a) Prove that union of two subgroups of a group G is a subgroup iff one is contained in the other. (8)

(b) Prove that a subgroup of a cyclic group is cyclic. (6)

(c) Show that the set of all 2×2 non singular matrices is a group under matrix multiplication. (6)

4. (a) Let H and K be two subgroups of a finite group G such that $|H| > \sqrt{|G|}$ and $|K| < \sqrt{|G|}$. Then prove that $H \cap K \neq \{e\}$. (10)

(b) State and prove fundamental theorem of homomorphism. (10)

5. (a) Show that any integral domain D can be imbedded in a field F and every element of F can be expressed as a quotient of two elements of D. (10)

(b) Let R be a commutative ring with identity. Show that R is a field iff R has no proper ideals. (10)

6. (a) Let R be an Euclidean domain and $a, b, c \in R$. Show that a / bc and $(a, b) = 1$ implies a/c (6)

(b) Show that a finite commutative ring R without zero divisors is a field. (8)

(c) If I is an ideal of a ring R with identity 1 and if $1 \in I$, then show that $I = R$. Also show that a field has no proper ideals. (6)

7. (a) Show that any vector space of dimension n over a field F is isomorphic to $V_n(F)$. (10)

(b) Let V be a finite dimensional inner product space. Let W be a subspace of V. Prove that V is the direct sum of W and $W^{\perp}$. (10)

8. (a) Let V be a finite dimensional vector space over a field F. Let A and B be subspaces of V. Prove that $\dim(A+B) = \dim A + \dim(A \cap B)$. (10)

(b) If V is a vector space over a field F and if $S, T \leq V$ then prove that $L(S \cup T) = L(S) + L(T)$ and $L(L(S)) = S$. (10)

DE–3607

Sub. Code
32

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the mathematical formulation of a linear programming problem and its matrix formulation. (6)

(b) Rewrite in standard form the following linear programming problem :

Minimize $z = 2x_1 + x_2 + 4x_3$

subject to the constraints :

$-2x_1 + 4x_2 \le 4$, $x_1 + 2x_2 + x_3 \ge 5$, $2x_1 + 3x_3 \le 2$, $x_1, x_2 \ge 0$ and x_3 unrestricted in sign. (6)

(c) Use graphical method to solve the L.P.P : (8)

Maximize $z = 2x_1 + 4x_2$

subject to the constraints :

$x_1 + 2x_2 \le 5$, $x_1 + x_2 \le 4$; and $x_2, x_2 \ge 0$.

2. (a) Write the dual of the L.P.P :

Minimize $z = 4x_1 + 6x_2 + 18x_3$

subject to the constraints :

$x_1 + 3x_2 \geq 3$, $x_2 + 2x_3 \geq 5$, and $x_j \geq 0$, $j = 1, 2, 3$. (5)

(b) Show that the dual of the dual is primal. (7)

(c) Use simplex method to solve the following L.P.P :

Maximize $z = 4x_1 + 10x_2$

subject to the constraints :

$2x_1 + x_2 \leq 50$, $2x_1 + 5x_2 \leq 100$, $2x_1 + 3x_2 \leq 90$; and $x_1, x_2 \geq 0$. (8)

3. (a) Use Big-M method to solve the following L.P.P :

Maximize $z = 6x_1 + 4x_2$

subject to the constraints :

$2x_1 + 3x_2 \leq 30$, $3x_1 + 2x_2 \leq 24$, $x_1 + x_2 \geq 3$; and $x_1 x_2 \geq 0$. (10)

(b) Enumerate the two-phase simplex method to solve the L.P.P. (10)

4. (a) What is integer linear programming? Explain the merits and demerits of "rounding off" a continuous optimal solution to an L.P.P to obtain an integer solution. (8)

(b) Using Gomory's cutting plane method to solve the following mixed integer programming problem :

Maximize $z = 4x_1 + 6x_2 + 2x_3$

subject to the constraints :

$4x_1 - 4x_2 \leq 5$, $-x_1 + 6x_2 \leq 5$, $-x_1 + x_2 + x_3 \leq 5$, $x_1, x_2, x_3 \geq 0$; x_1 and x_3 are integers. (12)

5. (a) What is a balanced transportation problem? (3)

(b) Obtain an initial feasible solution the following transportation problem using the north-west corner rule. (7)

	D	E	F	G	Available
A	11	13	17	14	250
B	16	18	14	10	300
C	21	24	13	10	400
Requirement	200	225	275	250	

(c) Find the optimal solution to the following transportation problem : (10)

	D_1	D_2	D_3	D_4	Available
O_1	22	27	16	18	30
O_2	12	17	20	51	40
O_3	22	28	12	32	53
Required	22	35	25	41	123

6. (a) (i) Give a brief outline of the procedure for solving a transportation problem.

(ii) Discuss the Hungarian method to solve the assignment problem. (10)

(b) Solve the following assignment problem. (10)

$$\begin{array}{c} \\ J_1 \\ J_2 \\ J_3 \\ J_4 \end{array} \begin{array}{c} \begin{array}{cccc} M_1 & M_2 & M_3 & M_4 \end{array} \\ \begin{pmatrix} 5 & 8 & 3 & 2 \\ 10 & 7 & 5 & 8 \\ 4 & 10 & 12 & 10 \\ 8 & 6 & 9 & 4 \end{pmatrix} \end{array}$$

7. (a) (i) Narrate the following terms.

Optimal strategies and Rectangular game.

(ii) For what value of λ, the game with following payoff matrix is strictly determinable? (10)

$$\begin{array}{cc} & \text{Player B} \\ & \begin{array}{ccc} B_1 & B_2 & B_3 \end{array} \\ \text{Player A} \begin{array}{c} A_1 \\ A_2 \\ A_3 \end{array} & \begin{pmatrix} \lambda & 6 & 2 \\ -1 & \lambda & -7 \\ -2 & 4 & \lambda \end{pmatrix} \end{array}$$

(b) Solve the following 2×4 game graphically : (10)

$$\begin{array}{cc} & \text{Player B} \\ & \begin{array}{cccc} B_1 & B_2 & B_3 & B_4 \end{array} \\ \text{Player A} \begin{array}{c} A_1 \\ A_2 \end{array} & \begin{pmatrix} 2 & 1 & 0 & -2 \\ 1 & 0 & 3 & 2 \end{pmatrix} \end{array}$$

8. (a) (i) What do you mean by a network?

(ii) Distinguish between PERT and CPM.

(iii) Enumerate the following terms :

Total float, Free float and independent float. (10)

(b) A project schedule has the following characteristics :

Activity	Time	Activity	Time
1 – 2	4	4 – 9	5
1 – 3	1	5 – 6	4
2 – 4	1	5 – 7	8
3 – 4	1	6 – 8	1
3 – 5	6	7 – 8	2
		8 – 10	5
		9 – 10	7

(i) Construct PERT network.

(ii) Find the critical path. (10)

DE-3608

Sub. Code
33

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Find the approximate value of the real root of $x\log_{10} x = 1.2$ by regular falsi method.

   (b) Prove that the order of convergence of the Newton Raphson method is atleast 2.

2. (a) Find the inverse of the given matrix by Gauss – Jordan method

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$$

   (b) Solve the following system of equations by Factorisation method.

$$2x - 6y + 8z = 24$$
$$5x + 4y - 3z = 2$$
$$3x + y + 2z = 16$$

3. (a) State and prove Newton – Gregory formula for forward interpolation.

   (b) Compute $f(3.6)$ from the following table.

x	0	1	2	3	4
$f(x)$	0.412	0.368	0.135	0.050	0.018

4. (a) Derive the formula for Laplace – Everelt's formula.

(b) Use Lagrange's formula to find the value of y at $x = 6$ from the following data

x	3	7	9	10
y	168	120	72	63

5. (a) Derive the formula for Trapezoidal rule.

(b) Evaluate $\int_0^{10} \frac{dx}{1+x^2}$ by using

(i) Trapezoidal rule

(ii) Simpson one third rule.

6. (a) Using Taylor's method solve $\frac{dy}{dx} = 1 + xy$ with $y_0 = 2$. Find

(i) $y(0.1)$

(ii) $y(0.2)$ and

(iii) $y(0.3)$.

(b) Using modified Euler method solve $\frac{dy}{dx} = 1 + xy$ with $y(0) = 2$. Find $y(0.1)$ $y(0.2)$ and $y(0.3)$.

7. (a) Compute $y(0.1)$ and $y(0.2)$ by Runge – Kutta method of 4th order for the differential equation $\frac{dy}{dx} = xy + y^2$, $y(0) = 1$.

(b) Find the value of $y(0.1)$ by Picard's method given $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$.

8. (a) Derive the formula for Milne's predictor and corrector method.

(b) Using Adam's Bashforth method find $y(0.4)$ given that $y' = 1 + xy$, $y(0) = 2$.

DE–3609

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DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) For any two complex numbers Z_1 and Z_2, prove that $|Z_1 + Z_2| \le |Z_1| + |Z_2|$ and $|Z_1 - Z_2| \le |Z_1| - |Z_2|$. (10)

   (b) State and prove Cauchy-Riemann equations in polar co-ordinates. (10)

2. (a) If $w = f(z) = u + iv$ and $u - v = e^x(\cos y - \sin y)$ find $f\,z)$ in terms of z. (10)

   (b) Prove that a power series represents a function that is analytic inside the circle of convergence of that series. (10)

3. (a) Find the necessary condition for $w = f(z)$ to represent a conformal mapping. (10)

   (b) Find the bilinear transformation which maps the points $Z = 0, 1, -1$ into $w = i, \infty, 0$. (10)

4. (a) Discuss the transformation $w = e^z$. (10)

(b) Show that the transformation $w = \frac{(5-4z)}{(4z-2)}$ maps of radius unity and centre $\frac{-1}{2}$. (10)

5. (a) Let f be analytic at all points inside and on a simple closed curve C. Then prove that $\int_C f(z)\,dz = 0$. (10)

(b) Find the Taylor series to represent $\frac{(z^2-1)}{(z+2)(z+3)}$ in $|z| < 2$. (10)

6. (a) State and prove fundamental theorem of algebra. (10)

(b) Evaluate $\int_C \frac{(3z-1)}{(z^3-z)}\,dz$ where C is $|z| = \frac{1}{2}$. (10)

7. (a) Expand $\frac{1}{(z^2-3z+2)}$ in Laurent's series valid in the region $1 < |z| < 2$. (10)

(b) State and prove Rouche's theorem. (10)

8. (a) Determine and classify the singular points of $f(z) = \frac{z}{(e^z-1)}$. (10)

(b) Evaluate using Cauchy's residue theorem $\int_C \frac{(z+1)}{(z^2+2z+4)}$ where C is a circle $|z+1+i| = 2$. (10)

DE-3610

Sub. Code
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DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, MAY 2018.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

Each questions carries equal marks.

1. (a) (i) Define disjunction and Biconditional statement with an example. (5)

   (ii) Construct the truth table for $(P \vee Q) \vee \neg P$. (5)

   (b) Show that $Q \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$ is a tautology. (10)

2. (a) Obtain the disjunctive normal form of a $P \wedge (P \to Q)$. (10)

   (b) Find the Principal disjunctive normal form of $P \to ((P \to Q) \wedge \neg(\neg Q \vee \neg P))$. (10)

3. (a) Show that $P \to S$ can be derived from the Premises $\neg P \vee Q,\ \neg Q \vee R,\ R \to S$. (10)

   (b) Show that $S \to \neg Q,\ S \vee R, \neg R, \neg R \leftrightarrow Q \Rightarrow \neg P$. (10)

4. (a) Show that $P(x) \wedge (x)Q(x) \Rightarrow (\exists x)(P(x) \wedge Q(x))$. (10)

   (b) Show that $(\forall x)(P(x)) \vee (\forall x)(Q(x)) \to (\forall x)(P(x) \vee Q(x))$ is logically valid. (10)

5. (a) Define the following with an example.

   (i) Digraph (2)

   (ii) Path of the graph (2)

   (iii) Cycle (2)

   (iv) Connected graph (2)

   (v) Path matrix. (2)

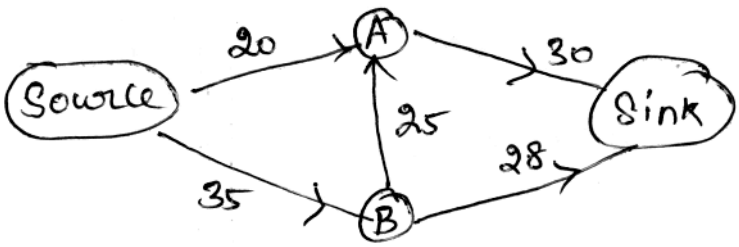
   (b) Prove that a tree with n vertices has $n-1$ edges. (10)

6. (a) How many different directed trees are there with three nodes? How many different ordered trees are there with three nodes? (10)

   (b) Let T be a full binary tree with n vertices. Prove that the number of leaves in T is $\frac{(n+1)}{2}$. (10)

7. (a) Prove that every hamiltonian graph is 2-connected. (10)

   (b) Find a maximum flow for the following network. (10)



8. (a) Prove that K_5 is non planar. (10)

   (b) Prove that if G is a simple planar graph with atleast three vertices $e(G) \le 3n(G) - 6$. (10)